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Supplemental information

Detection and quantification of integrated vector copy number by multiplex droplet digital PCR in dual-transduced CAR T cells

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Supplemental Figures

Temperature Gradient

Well	Tm (^oC)
A	65
B	64.4
C	63.1
D	61.2
E	59
F	57.1
G	55.8
H	55

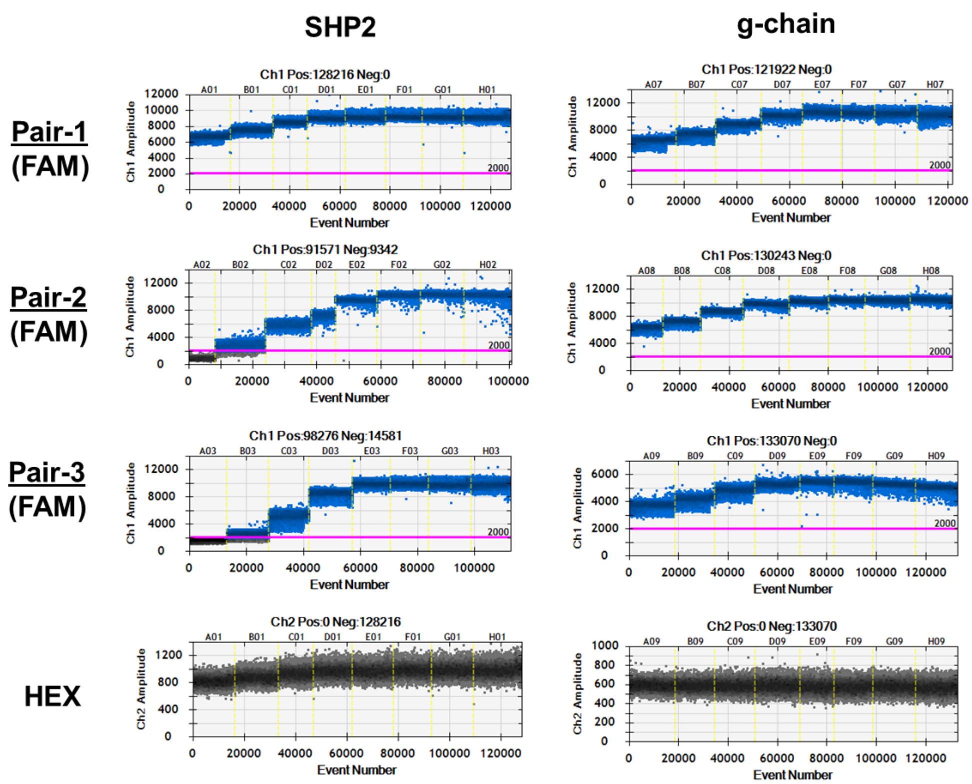

Figure S1. Determination of the optimal annealing temperature for ddPCR assay.

The annealing temperature (Tm) was optimized using a thermal gradient (55°C - 65°C) with all three pairs of primers-probe for each target (SHP2 and g-chain) at the same final concentration. 59 °C was selected as the optimal annealing temperature based on the highest temperature that produced the best separation of negative and positive droplets in FAM channel.

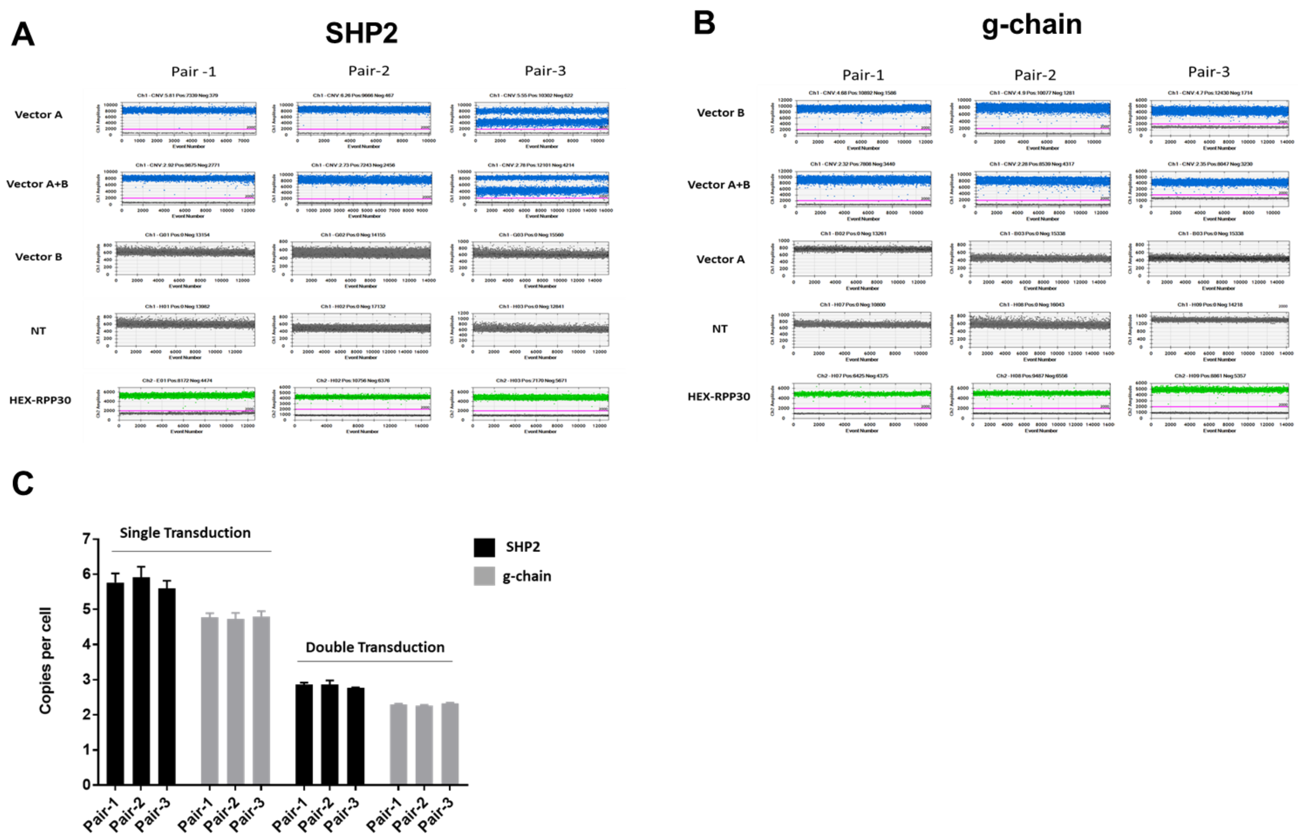

Figure S2. Analysis of primer-probes for SHP2 and g-chain.

Three specifically designed pairs of primers-probe were used to detect and quantify SHP2 (A) or g-chain (B) in FAM channel using Vector A single-transduced cells, Vector B single-transduced cells, Vector A+B double-transduced cells, and non-transduced cells (NT). RPP30 was measured in the HEX channel. (C) Comparison of SHP2 and g-chain VCN (copies per cell) quantification using each of the three pairs of primer-probe. Bars indicate mean values with SDs (n=3).

Supplemental Tables

Table S1. Determination of assay linear range, precision, LoD, LoQ.

RPP30													
gDNA (ng)	**Rep1**	**Rep2**	**Rep3**	**Rep4**	**Rep5**	**Rep6**	**Rep7**	**Rep8**	**Rep9**	**Rep10**	**Mean**	**SD**	**CV%**
100	19918	18423	17963	18170	18929	18584	18354	17756	18814	18538	18545	602	3.2
100	20861	20677	19941	19596	18377	20309	19159	19251	18354	18814	19534	901	4.6
100	21183	20792	20585	20033	21459	19458	20033	20102	19918	19274	20284	710	3.5
100	22011	21827	19412	20769	20286	20286	19596	19435	19274	18768	20166	1092	5.4
100	21781	20125	21160	20654	19803	20999	20033	19826	19964	19688	20403	707	3.5
100	20631	20930	22425	21206	20677	21275	20401	19872	20424	21068	20891	687	3.3
100	20792	21022	22264	23851	22080	20585	20240	19412	21068	19918	21123	1301	6.2
100	22126	21137	21643	20815	21137	20815	19389	19228	19274	19826	20539	1041	5.1
100	23598	22356	22402	22057	21413	20424	20792	20723	20378	19964	21411	1158	5.4
100	22356	20217	20861	22471	22586	21965	20976	20953	20654	20378	21342	908	4.2
100	22839	21206	22632	18975	21942	21597	20240	20424	20838	21114	21181	1157	5.5
0	0	0	0	0	0	0	0	0	0	0	0	0	N/A
SHP2													
Expected	**Rep1**	**Rep2**	**Rep3**	**Rep4**	**Rep5**	**Rep6**	**Rep7**	**Rep8**	**Rep9**	**Rep10**	**Mean**	**SD**	**CV%**
500000	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
100000	99560	108680	109060	110580	104500	110200	107540	104120	101840	104880	106096	3701	3.5
50000	52866	51414	51150	51678	49698	48609	50325	49599	48444	47388	50117	1692	3.4
5000	5313	5175	4761	4830	4830	4807	4646	4554	5014	4738	4867	235	4.8
2500	2254	2323	2323	2323	2484	2346	2139	2093	2208	2438	2293	123	5.4
500	580	499	561	501	543	541	531	462	455	474	515	43	8.3
200	202	186	156	207	193	154	191	173	179	189	183	18	9.7
100	92	76	109	99	102	89	96	89	106	102	96	9	10.2
50	60	55	44	51	41	58	58	39	53	48	51	7	14.4
10	10	9	9	9	10	6	11	9	7	8	9	1	15.4
5	6	9	8	6	11	12	8	5	4	4	7	3	39.4
0	0	0	0	0	0	0	0	0	0	0	0	0	N/A
g-chain													
Expected	**Rep1**	**Rep2**	**Rep3**	**Rep4**	**Rep5**	**Rep6**	**Rep7**	**Rep8**	**Rep9**	**Rep10**	**Mean**	**SD**	**CV%**
500000	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
100000	99590	97290	100510	105570	105110	105110	106260	103500	96140	97980	101706	3835	3.8
50000	52050	46950	47400	48900	51750	48000	46350	46500	48150	47700	48375	2012	4.2
5000	4644	6264	4230	4716	4806	5454	4158	4734	5688	5364	5006	669	13.4
2500	2277	2783	2875	2737	2990	2990	2369	2760	2668	2484	2693	246	9.1
500	580	375	382	439	426	391	506	414	455	497	446	65	14.5
200	186	258	235	253	225	230	175	193	260	244	226	31	13.7
100	86	92	109	76	119	119	109	106	92	86	99	15	15.0
50	64	58	48	39	53	48	46	51	44	41	49	8	15.5
10	7	12	7	12	9	8	10	6	6	8	8	2	23.8
5	2	6	2	6	11	2	4	13	3	5	5	4	71.4
0	0	0	0	0	0	0	0	0	0	0	0	0	NA

Table S2. Analysis of gDNA interference and dynamic range.

RPP30													
gDNA (ng)	**Rep1**	**Rep2**	**Rep3**	**Rep4**	**Rep5**	**Rep6**	**Rep7**	**Rep8**	**Rep9**	**Rep10**	**Mean**	**SD**	**CV%**
400	74400	78000	73440	73680	72480	72000	69120	70800	72240	73440	72960	2346	3.2
200	39840	38400	36768	38664	34248	35640	36480	34584	36432	37440	36850	1785	4.8
100	18336	17136	17496	16704	18672	17904	16608	15984	17232	16752	17282	834	4.8
50	8400	8376	8520	8424	9096	8160	7920	7584	7872	8064	8242	421	5.1
10	1716	1630	1906	1694	1615	1620	1474	1450	1603	1670	1638	128	7.8
5	785	1008	986	734	878	838	838	830	859	974	873	90	10.3
2.5	463	362	331	394	422	362	410	437	374	403	396	40	10.0
1.25	250	218	221	173	254	221	175	204	211	230	216	27	12.5
0.25	41	72	55	53	38	58	50	48	67	62	54	11	19.8
0.05	7	0	5	5	5	2	4	10	3	9	5	3	60.5
0.01	0	0	9	3	0	0	0	4	0	1	2	3	172.5
0	0	0	0	0	0	0	0	0	0	0	0	0	N/A
SHP2													
gDNA (ng)	**Rep1**	**Rep2**	**Rep3**	**Rep4**	**Rep5**	**Rep6**	**Rep7**	**Rep8**	**Rep9**	**Rep10**	**Mean**	**SD**	**CV%**
400	77040	78720	72960	76320	74160	73920	69600	71520	72240	72720	73920	2753	3.7
200	72000	67440	66000	70560	64080	64560	65280	63600	72240	73440	67920	3776	5.6
100	77760	72720	77040	73680	80640	76320	71520	67920	73440	76320	74736	3621	4.8
50	72960	72000	71520	75360	73680	72480	66960	62496	73920	71040	71242	3802	5.3
10	68400	70320	77040	71280	65520	66240	63120	67200	72240	75840	69720	4492	6.4
5	58152	69120	67680	62640	68880	64320	63360	55728	71040	73440	65436	5643	8.6
2.5	70080	63840	63120	68880	68160	70080	81120	80400	73440	76320	71544	6233	8.7
1.25	71520	65040	61800	57960	61344	57048	76800	77088	77040	81120	68676	9031	13.1
0.25	66570	75930	75300	75120	74520	72480	69630	68280	76200	82800	73683	4685	6.4
0.05	70100	65700	84950	82000	88850	77250	65650	62950	69400	78300	74515	8993	12.1
0.01	77700	69755	67165	75845	71610	69615	81305	66150	76300	70280	72573	4950	6.8
0	78430	72875	71280	81840	89540	62975	72600	83930	64185	76395	75405	8404	11.1
g-chain													
gDNA (ng)	**Rep1**	**Rep2**	**Rep3**	**Rep4**	**Rep5**	**Rep6**	**Rep7**	**Rep8**	**Rep9**	**Rep10**	**Mean**	**SD**	**CV%**
400	123360	127680	121680	126000	122640	121680	123120	120000	124320	121440	123192	2297	1.9
200	142560	140640	134640	135360	123840	126000	121440	122640	121920	120240	128928	8509	6.6
100	118080	108720	112560	111360	115200	114240	118080	108720	122880	126720	115656	5896	5.1
50	120240	115920	119520	115920	119760	126480	124320	107760	117120	120720	118776	5142	4.3
10	110400	108720	118560	112080	103680	107760	116640	106560	122880	121920	112920	6692	5.9
5	102480	121680	109680	115200	119760	114240	103680	97920	109440	119280	111336	8082	7.3
2.5	123600	97200	115920	127440	98640	101760	109200	115920	119520	121440	113064	10792	9.5
1.25	97680	103680	101040	95520	100080	97920	114000	110640	125520	123840	106992	10977	10.3
0.25	100500	110400	103800	102900	101700	105300	98100	98400	121500	115800	105840	7741	7.3
0.05	101880	93015	112725	111600	96795	107550	108450	102600	102420	107010	104405	6283	6.0
0.01	98100	96000	118500	92400	95610	86400	90240	112800	89400	93690	97314	10344	10.6
0	118900	109900	126650	110900	100150	100450	108000	99900	117250	100500	109260	9362	8.6